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Delayed Fertilization and Preimplantation Loss in Senescent Golden Hamsters[1, 2]

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Ova were collected from young virgin (120 animals, 3-5 months of age) and senescent multiparous (172 animals, 14-17 months of age) female golden hamsters from 1 h through 3½ days postovulation. The majority of ova recovered from aged hamsters mated with young males exhibited a 2 to 5 h delay in fertilization. This difference was not attributed to delayed ovulation in the older female, but rather to prolonged penetration of the zona pellucida and vitellus by the spermatozoon, extending the time of fertilization. Approximately 40 percent of the ova ovulated by senescent females were non-viable (unfertile, abnormal, and degenerating) at the time of implantation. Many of the non-viable ova could have resulted directly from the aging of the ovum before fertilization, while others appear to be inherently defective at the time of ovulation.

Preimplantation loss in the senescent golden hamster is the most important factor in the declining litter size.

One of the characteristics of aging in female laboratory mammals is a progressive decline in litter size after they have borne several optimum litters (Ingram *et al.*, 1958; Soderwall *et al.*, 1960; Biggers *et al.*, 1962). The decline is not caused by a depletion of oocytes, since some females continue to ovulate adequate numbers of ova even after the female fails to reproduce any offspring (Harman and Talbert, 1970; Jones, 1970). The aging uterus is generally considered as the organ most responsible for a reduction of offspring (Biggers, 1969; Finn, 1970; Adams, 1970; Spilman *et al.*, 1972), although the viability of a larger number of ova in older females may also be impaired (Blaha, 1964; Henderson and Edwards, 1968; Yamamoto *et al.*, 1973).

The female golden hamster exhibits a sharp decline in the number of young born after 14 months of age (Soderwall *et al.*, 1960). Part of this reduction results from twice as many resorptions in senescent females compared with young females (Thorneycroft and Soderwall, 1969), while another important factor appears to be preimplantation loss (Parkening and Soderwall, 1973b). The gestational period, extended in senescent hamsters, is characterized by approximately a 12 h delay in the development of preimplantation and postimplantation embryos when compared with embryos from young females at the same age of pregnancy (Connors, 1969; Connors *et al.*, 1972; Parkening and Soderwall, 1973a, 1973b). The senescent female uterus is similarly delayed in its ability to form decidual cells during normal pregnancy (Parkening and Soderwall, 1973a) and to accept transferred blastocysts from young females during pseudopregnancy (Stockton *et al.*, 1973). This suggests that the synchrony between the uterus and the implanting blastocyst may not be disturbed in older hamsters even though it is different from younger animals.

The ova of laboratory mammals remain fertilizable longer than their ability to develop into normal embryos. The presence of large

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numbers of non-viable ova and the delay in embryonic development in senescent females could both result from delayed fertilization. To investigate this possibility, a comparison of the early stages of fertilization and development in young and senescent hamsters was conducted. A preliminary report of this study appeared earlier (Parkening and Soderwall, 1974).

MATERIALS AND METHODS

Animals

Golden hamsters (*Mesocricetus auratus*) housed under constant conditions of lighting (13 h of light—0600 to 1900) and temperature (21–23° C) were fed lettuce and laboratory animal chow daily. All of the young male and female hamsters and a majority of the senescent females were born and raised in the Oregon colony; however, 72 females were purchased as retired breeders (10 months of age) from Hilltop Lab Animals (Scottdale, Pennsylvania) and Engle Laboratory Animals (Farmersburg, Indiana) and maintained until 14 to 17 months of age.

Data were collected uniformly over a 14-month period utilizing 120 young nuliparous females (3–5 months of age) and 172 senescent multiparous females (14–17 months of age). Each female exhibited 2 to 5 normal 4 day estrous cycles, based on vaginal discharge (Orsini, 1961), before being bred with males (3–6 months of age) of proven fertility. Animals were placed together in the early evening and observed for mating. As soon as a female exhibited lordosis, this was considered the beginning of behavorial estrus and ovulation was assumed to occur 8 h post-estrus (Harvey *et al.*, 1961). The age of all fertilized and unfertilized ova was based on this presumed time of ovulation.

Recovery and Examination of Ova

Females were killed by cervical dislocation every hour from 1 through 12 h and at 1, 1½, 2, 2½, 3, and 3½ days postovulation. For the recovery of ova after 1 h, the ovary and oviduct were removed with the ovarian sac intact and placed in Hanks' solution (1–2 ml) where the sac was dissected away using a stereoscopic microscope. This permitted the counting of ova, freshly ovulated, which had not yet reached the ampulla of the oviduct. The oviduct was then separated from the ovary and carefully dissected. Ova recovered at other time intervals were obtained by removing the oviduct and dissecting it in Hanks' solution or by removing and flushing each uterine horn two or three times with a hypodermic syringe containing Hanks' solution. Several drops of 0.1 percent hyaluronidase (dissolved in Hank's solution) were added for the dispersion of cumulus cells adhering to ova from 1 through 12 h postovulation. The ova were transferred to a glass slide containing two drops of 2.5 percent glutaraldehyde (phosphate buffered, pH 7.3) and mounted with a cover slip supported by Vaseline at its corners to allow for free movement of the ova within the solution. They were examined with phase and differential interference contrast microscopy (Nomarski optics).

Approximately one-half of the normal appearing ova recovered from females at 3 through 12 h postovulation were randomly selected and mounted in Hanks' solution, compressed with a cover slip and left in 10 percent formalin overnight (Iwamatsu and Chang, 1972). After fixation the ova were stained with 0.25 percent lacmoid in 45 percent acetic acid (Yanagimachi and Chang, 1961) and photographed with phase microscopy.

The criteria for classifying an ovum as fertile were dependent upon a spermatozoon penetrating the zona pellucida and attaching to the vitellus together with the extrusion of the second polar body. A swollen spermatozoan head and the appearance of male and female pronuclei were an added indication of fertilization.

Twenty-one senescent hamsters used in a separate study were placed with males and observed for the onset of behavioral estrus, but were removed prior to mating. These females were placed with males 4 h later, so that they copulated approximately 4 h before ovulation. Ova were collected from these animals, compressed, fixed, and stained as described above.

Examination of Spermatozoa

One hundred spermatozoa were recovered from dissected oviducts of young and senescent females from 3 through 8 h postovulation for an examination of the acrosome reaction.

To determine how fast spermatozoa were capable of migrating through the female reproductive tract, six young and six senescent hamsters were allowed to mate with two males for 1 h from the onset of behavioral estrus. The animals were then anesthetized, perfused with 2.5 percent glutaraldehyde (phosphate buffered, pH 7.3 at 4° C) and the oviducts removed and fixed in glutaraldehyde for an additional 2 h. After postfixation in 2.0 percent osmium tetroxide for 2 h, the oviducts were dehydrated and embedded in Epon-Araldite. Serial sections (3–5 μm) were cut dry with glass knives on a Porter-Blum MT-1 ultramicrotome, floated on water on glass slides, and dried on a slider warmer. The sections were examined unstained using phase microscopy. Plastic was utilized as an embedding medium to prevent shrinkage artifacts commonly associated with standard paraffin embedding techniques.

Artificial Insemination

The fertilizing life of hamster spermatozoa was determined by observing 20 young females on three separate occasions for the beginning of behavorial estrus. Once the approximate time these females exhibited estrus was known, they were anesthetized with ether at 8, 14, 15, and 16 h prior to the expected time of ovulation. The spermatozoan suspension was prepared by mincing a cauda epididymis in 4 ml of Hanks' solution. The uterine horns were exposed through dorso-lateral incisions and 0.2 ml of spermatozoa (4.5×10^7/ml) were injected into the lumen of each horn by means of a 24-gauge needle attached to a tuberculin syringe. The procedure was identical to that described by Miyamoto and Chang (1972) except the Hanks' solution was warmed to 37° C before mincing the epididymis. Three animals were inseminated at one time, and the entire process, from mincing the epididymis to the injection of spermatozoa in all horns of the three animals, lasted approximately 5 min. Ova were obtained either between 8 and 9 h or between 24 and 48 h after the expected time of ovulation. They were examined for penetration of the zona pellucida by a spermatozoon, incorporation of the spermatozoan head, extrusion of the second polar body, and cleavage.

Statistics

The statistical method for comparing differences between means was Student's t test (Sokal and Rohlf, 1969).

RESULTS

Ovulation

The mean number of ova collected 1 h after the expected time of ovulation was 13.8 ova for young hamsters and 10.5 for senescent hamsters (Table 1). The difference of 3.3 ova was statistically significant ($p < 0.05$). An even greater difference existed when comparing the number of normal ova in the two groups. Of 84 ova examined from eight aged hamsters, 10 were either abnormal or degenerating, leaving a mean of 9.3 ova compared with 13.8 ova for younger animals. None of the ova from young females was visibly defective at this stage of development. The ova from two senescent females were not included in the above data because an abnormally large number of ova (78) were found, of which most (67) were in some stage of degeneration. Of the 18 time intervals studied, in all but four time periods, more than 20 ova were recovered from one to four animals at each interval. Animals exhibiting this trait of excessive ova were restricted to aged hamsters, except for two young animals. Of the 31 females displaying this condition, the majority contained an average number of normal appearing ova, with the remaining ova in some state of degeneration. A histological examination of several ovaries of these animals revealed that the number of corpora lutea corresponded to the number of normal appearing ova. For this reason these animals and their degenerating ova have been eliminated from Table 1. The number of normal ova from all young and senescent animals was included in Table 1 for a more representative sample of the mean number of normal ova between the two groups.

At 2 h postovulation an average of 12.8 ova were recovered from aged females, which was not significantly different statistically from the average of 13.8 ova collected from young females at 1 h postovulation. However, the mean number of normal ova (those not visibly abnormal or degenerating) was 10.3 for the senescent group, which was significantly different ($p < 0.01$) when compared with normal ova from young females at 1 h.

Fertilization

The majority of ova from senescent hamsters exhibit a delay in fertilization varying from 2 to 5 h beyond that of young hamsters (Figs. 1–6). This is based upon the total number of ova in which the zona pellucida or vitellus was penetrated by spermatozoa, the extrusion of the second polar body, and the formation of pronuclei (Table 2). At 4 and 5 h postovulation 90.1 percent and 100 percent of the ova from young animals had the zona pellucida penetrated by spermatozoa, whereas only 37.3 percent and 40.9 percent of the ova had been penetrated during the same time period in aged animals. In a similar manner the percentage of these ova with a spermatozoon in close contact with the vitellus and the subsequent formation of a second polar body was 77.5 percent and 94.3 percent for young females, compared to 25.3 percent and 29.6 percent after 4 and 5 h in senescent females. Not until 8 h after ovulation did 94.4 percent of the ova from senescent animals contain spermatozoa and of these only 78.9 percent had extruded a second polar body. Most of the ova (73.7 percent) containing a second polar body at this time period had formed both a male and female pronucleus (Fig. 6). Seventy-five percent of the ova from young hamsters contained a male and female pronucleus at 5 h postovulation (Fig. 5). At 11 and 12 h postovulation, 93.1 percent and 91.1 percent of the ova from senescent animals contained both pronuclei (Fig. 9). Of the number of senescent animals examined at 9 and 10 h postovulation, 80.3 percent and 72.3 percent of the ova had the zona pellucida penetrated by a spermatozoon. Both are relatively low percentages considering that 94.4 percent and 94.1 percent were penetrated at 8 and 11 h, and may reflect a variation in the fertilizability of the ova from senescent females.

There appeared to be a delay in the extrusion of the second polar body in some ova from senescent females, once the vitellus had been penetrated by a spermatozoon. All of the ova from young females in which the vitellus was penetrated 5 h after ovulation, had expelled the second polar body. In contrast, similarly penetrated ova from senescent females, 5 to 12 h after ovulation, had chromosomes of the second polar body at metaphase (4 ova) or in the process of dividing (39 ova). Twelve of these ova had prominently swollen spermatozoan heads, which might imply a lag in the formation of the second polar body. This could result in an unbalanced chromosomal distribution leading to increased zygote mortality in the aged female. One ovum from a senescent female at 12 h postovulation which exhibited two polar bodies had a swollen spermatozoan head in the vitellus, but no pronuclei had formed (Fig. 12).

Preimplantation Loss

A comparison of the mean number of total ova and normal ova from young and senescent hamsters is given in Table 1. Although the number of ova remains relatively stable in young females from ovulation (13.8) to the time of implantation (11.1), the difference of 2.7 ova is statistically significant ($p < 0.02$) and indicates some preimplantation loss in young animals. In the senescent female the number of ova from 2 h (10.3) to 2 days (9.2) remains relatively constant, but drops at 2½ days (6.8) and remains at this level through 3½ days of pregnancy. The difference between the mean number of ova from senescent animals at 1 h and 3½ days is not significant ($p < 0.1$), but the difference between those at 2 h and 3½ days is significant ($p < 0.02$). The latter appears to be a more valid comparison because of the average number of ova obtained during various time intervals from 2 h to 2 days. A comparison of the mean number of ova in young animals at 3 days of pregnancy (11.1) with those of senescent animals at 3½ days of pregnancy (6.9) is statistically highly significant ($p < 0.001$). The difference of 4.2 ova between the two groups emphasizes how important preimplantation loss is in senescent hamsters.

Of the normal appearing ova from young and aged females through 12 h postovulation, one ovum was abnormal from a young female and 12 ova were defective from senescent females. The abnormalities consisted of dispermy, trispermy, fragmented pronuclei and androgenesis (Figs. 7, 8, 10, 11). Of the 12 abnormal ova from aged females, seven occurred at 12 h accounting for 8.1 percent of the ova previously considered normal before staining with lacmoid. Ova acquired from both groups of hamsters beyond 12 h of ovulation were not stained with lacmoid. From these animals seven ova from young females and 31 ova from senescent females were considered abnormal. All the ova from young females were misshapen zygotes, having undergone cleavage. Most of the ova from aged hamsters were also of this type; however, 10 ova having unusually thin zonae pellucidae were collected from three females, all of which contained other ova with a normal appearing zona pellucida.

There were relatively few degenerating ova (those in an advanced state of fragmentation) from young animals examined at various stages of development prior to implantation (0.0 percent to 1.7 percent). No degenerating ova were present in senescent females at 8 h postovulation, but the number of such ova at the other time periods

TABLE 1

NUMBER OF NORMAL AND NON-VIABLE OVA AND EMBRYOS FROM YOUNG AND SENESCENT HAMSTERS AT VARIOUS TIME INTERVALS BETWEEN OVULATION AND IMPLANTATION

Hours post-ovulation	No. females		Mean ± standard error of ova & embryos*		No. of non-viable ova* (%)		No. of normal ova & embryos		Mean ± standard error of normal ova & embryos	
	Y	S	Y	S	Y	S	Y	S	Y	S
1	6	10	13.8 ± 0.91	10.5 ± 1.12	0 (0.0)	11 (10.6)	83	93	13.8 ± 0.91	9.3 ± 0.83
2	—	7	—	12.8 ± 2.25	—	16 (18.2)	—	72	—	10.3 ± 0.72
3	7	7	13.7 ± 0.89	12.0 ± 1.41	1 (1.0)	7 (8.9)	95	72	13.6 ± 0.95	10.3 ± 0.68
4	9	7	12.9 ± 1.18	13.0 ± 1.43	5 (4.3)	10 (10.8)	111	83	12.3 ± 0.97	11.9 ± 1.06
5	7	8	13.1 ± 1.28	12.0 ± 1.97	4 (4.3)	5 (5.6)	88	88	12.6 ± 1.11	11.0 ± 1.41
6	—	12	—	12.9 ± 0.61	—	11 (7.5)	—	135	—	11.3 ± 1.18
7	—	11	—	11.6 ± 0.92	—	4 (3.1)	—	123	—	11.2 ± 0.78
8	7	8	14.4 ± 0.20	11.6 ± 0.78	0 (0.0)	0 (0.0)	101	90	14.4 ± 0.20	11.3 ± 0.75
9	—	6	—	14.3 ± 1.37	—	10 (12.3)	—	76	—	12.7 ± 1.15
10	5	9	12.0 ± 1.26	10.7 ± 2.42	1 (1.7)	5 (5.1)	59	94	11.8 ± 1.36	10.4 ± 1.57[a]
11	6	6	12.2 ± 1.01	11.5 ± 1.26	13 (17.8)	5 (7.2)	60	64	10.0 ± 2.19[b]	10.7 ± 1.28
12	10	12	12.9 ± 0.56	12.6 ± 0.87	6 (4.7)	9 (6.6)	123	128	12.3 ± 0.82	10.7 ± 0.74
24	11	10	12.3 ± 0.68	10.9 ± 0.78	2 (1.5)	10 (9.2)	133	99	12.1 ± 0.69	9.9 ± 0.84
36	14	13	12.4 ± 0.61	11.7 ± 0.75	4 (2.6)	22 (15.6)	150	119	10.7 ± 1.13	9.2 ± 0.62
48	12	10	12.9 ± 1.67	11.7 ± 0.97	4 (2.9)	26 (22.0)	135	92	11.3 ± 1.28	9.2 ± 1.33

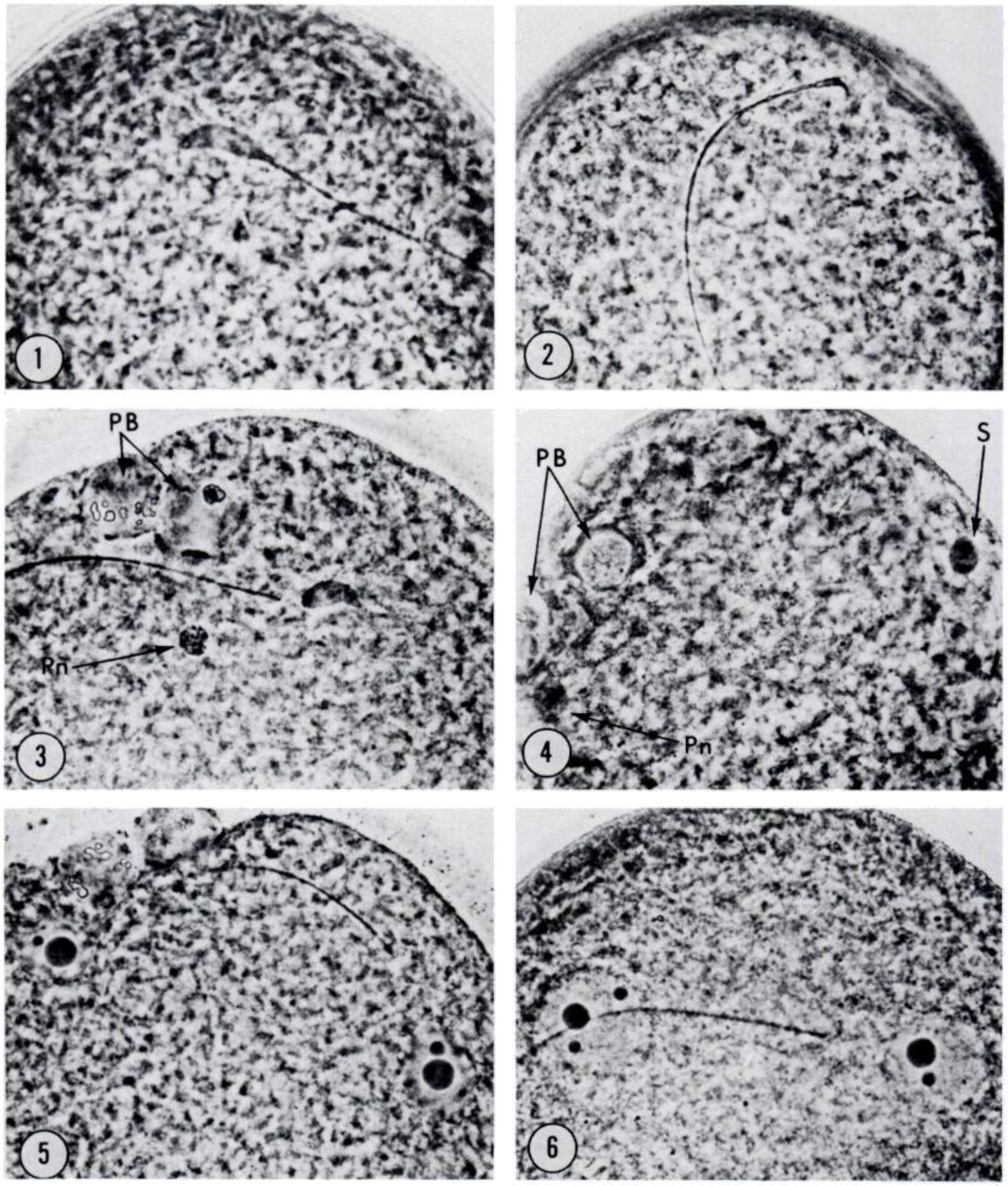
60	14	11	11.1 ± 0.55	9.6 ± 1.02	2 (1.3)	31 (29.2)	154	75	11.0 ± 0.58	6.8 ± 1.03
72	12	11	11.5 ± 0.54	10.9 ± 1.55	5 (3.6)	38 (34.9)	133	71	11.1 ± 0.63	7.1 ± 0.77
84	—	14	—	9.5 ± 0.64	—	29 (24.6)	—	89	—	6.9 ± 0.87
Total	120	172								

Y = Young, S = Senescent.
* Data excludes animals exhibiting excessive ova.
[a] No ova were found in one animal.
[b] All ova were infertile in one animal.

ranged from four (3.1 percent) to 37 (34.0 percent). Some of these ova acquired from aged animals during the first 12 h were undoubtedly defective at the time of ovulation. The number of degenerating ova from senescent animals at 2½ (17.9 percent), 3 (34.0 percent), and 3½ (16.9 percent) days of pregnancy was especially large in comparison with those examined at earlier periods. Many of the deteriorating ova recovered at these later stages had undergone cellular divisions and may have been reflections of the delay in fertilization rather than resulting from an intrinsic inferiority of the eggs.

Nineteen ova from young females and 27 ova from senescent females were recorded as infertile from 11 h until the time of implantation (ova included under non-viable ova in Table 1). A young female (11 h postovulation) ovulated 11 ova, all infertile, and a senescent female (10 h postovulation) did not yield any ova. The latter animal contained ripe follicles on both ovaries, but they had failed to rupture. In only two instances were gross anatomical obstructions visible which would prevent spermatozoa from passing to the oviduct. Two senescent animals each displayed a tumor in one horn, resulting in 10 infertile ova in one animal (1½ days pregnant) and five infertile ova in another animal (2 days pregnant).

Most of the fertilized ova recovered from young and senescent hamsters at 1 and 1½ days of pregnancy had cleaved or were in the process of cleaving. At 2 days a developmental lag, based upon the number of blastomeres, became apparent in zygotes examined from senescent animals. Of the total number of zygotes recovered from young and senescent animals at 2 days of pregnancy, 73.2 percent from young animals were at the 4-cell stage, while only 28.0 percent from old animals were at this stage of development. All embryos from young hamsters at 2½ days of pregnancy were within the uterine horns, predominantly in the 9-to 12-cell stage (77.6 percent), whereas embryos from senescent hamsters remained in the oviducts, with several at the 4-cell stage (17.9 percent), and the



All ova were stained with lacmoid and photographed with phase optics.

FIG. 1. An ovum from a young hamster 3 h after ovulation. A spermatozoon has entered the vitellus of the egg and its head has begun to swell. ×600.

FIG. 2. An ovum acquired from an aged hamster at 6 h postovulation. A spermatozoon has penetrated the zona pellucida and has made contact with the vitellus of the ovum, but the spermatozoan head has not been incorporated into the vitellus. ×480.

FIG. 3. A fertilized ovum from a young hamster at 5 h postovulation. The spermatozoan head is beginning its transformation into the male pronucleus while the female pronucleus (Pn) has just formed. Both polar bodies (PB) are visible with condensed chromatin. ×500.

FIG. 4. An ovum from a senescent hamster 9 h after ovulation which is comparable in development to the ovum of the young hamster in fig. 3. The swollen spermatozoan head (S), female pronucleus (Pn), and polar bodies (PB) are visible. ×500.

FIG. 5. A fertilized ovum from a young hamster 5 h after ovulation. Both male and female pronuclei are fully formed while the spermatozoan tail remains visible against the vitellus. ×500.

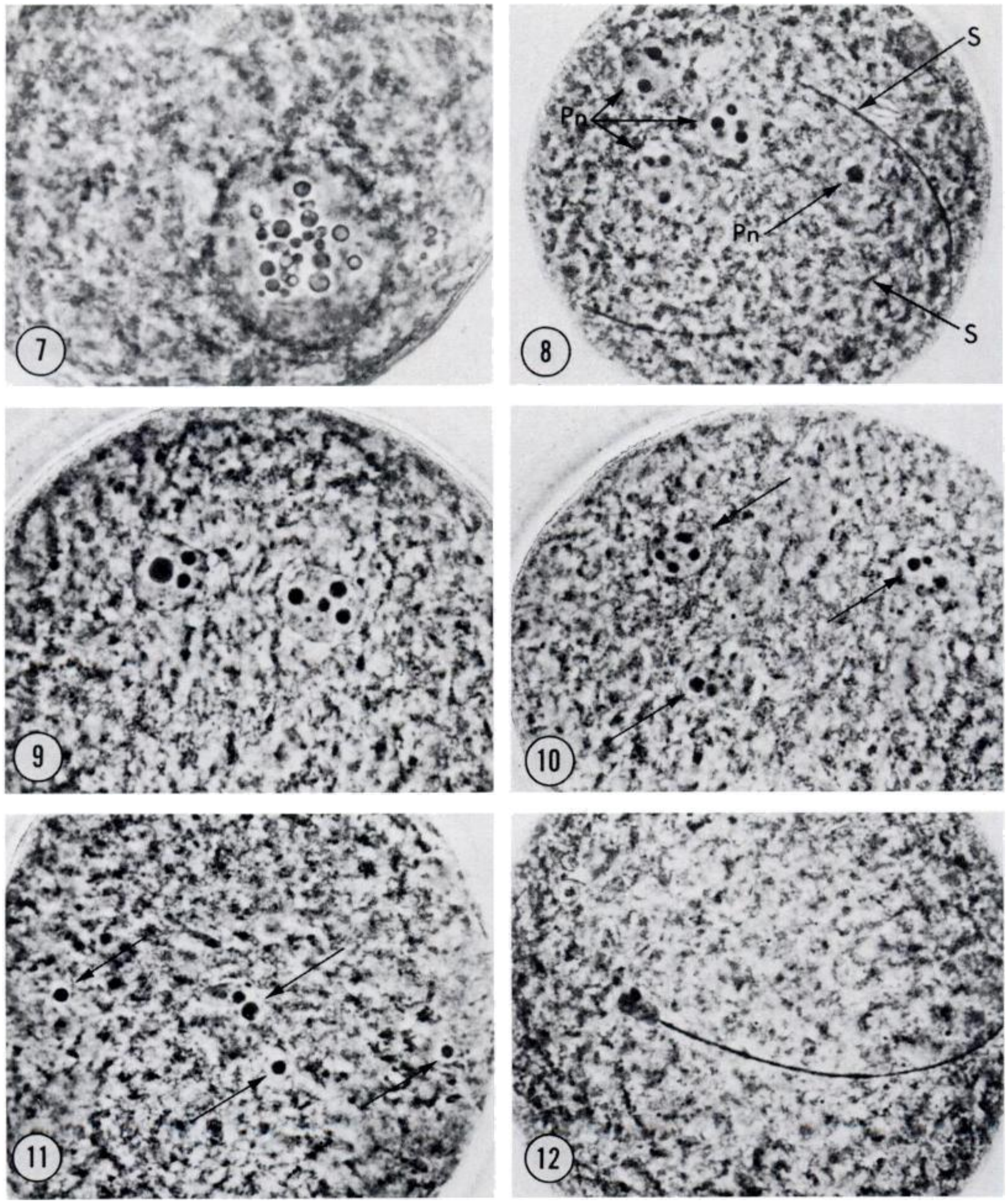
FIG. 6. A fertilized ovum from a senescent hamster at 10 h postovulation. The developmental stage of the ovum is comparable to that from the young animal at 5 h postovulation (fig. 5). ×500.

TABLE 2
NUMBER OF OVA PENETRATED BY SPERMATOZOA AND THEIR STAGE OF DEVELOPMENT FROM YOUNG AND SENESCENT HAMSTERS

Hours post-ovulation		No. of females with penetrated ova[a] (% of total)	Total no. of ova[b]	No. of ova penetrated (% of total)	No. of ova with 2nd polar body (% of total)	Total no. of ova stained with lacmoid	No. of ova with male & female pronuclei (% of total)
2	Y	—	—	—	—	—	—
	S	0 (0.0)	72	0 (0.0)	0 (0.0)	0	0 (0.0)
3	Y	5 (71.4)	95	42 (44.2)	28 (29.5)	54	0 (0.0)
	S	3 (42.9)	72	12 (16.7)	6 (8.3)	43	0 (0.0)
4	Y	9 (100)	111	100 (90.1)	86 (77.5)	54	0 (0.0)
	S	6 (85.7)	83	31 (37.3)	21 (25.3)	49	0 (0.0)
5	Y	7 (100)	88	88 (100)	83 (94.3)	40	30 (75.0)
	S	7 (87.5)	88	36 (40.9)	26 (29.6)	62	1 (1.6)
6	Y	—	—	—	—	—	—
	S	11 (91.7)	135	69 (51.1)	49 (36.3)	67	14 (20.9)
7	Y	—	—	—	—	—	—
	S	11 (100)	123	96 (78.0)	81 (65.9)	53	20 (41.5)
8	Y	7 (100)	101	101 (100)	101 (100)	62	62 (100)
	S	8 (100)	90	85 (94.4)	71 (78.9)	57	42 (73.7)
9	Y	—	—	—	—	—	—
	S	5 (83.3)	76	61 (80.3)	60 (78.9)	26	21 (80.8)
10	Y	5 (100)	59	59 (100)	59 (100)	36	36 (100)
	S	7 (77.8)	94	68 (72.3)	65 (69.1)	30	30 (100)
11	Y	6 (100)	71[c]	60 (84.5)	60 (84.5)	39	38 (97.4)
	S	6 (100)	68[c]	64 (94.1)	62 (91.2)	29	27 (93.1)
12	Y	10 (100)	128[c]	123 (96.1)	123 (96.1)	65	62 (95.4)
	S	12 (100)	128	128 (100)	126 (98.4)	90	82 (91.1)

Y = Young, S = Senescent.
[a] An ovum was penetrated once a spermatozoon passed through the zona pellucida or adhered to the vitellus.
[b] Includes normal appearing ova from females ovulating excessive numbers of ova.
[c] Some ova were considered infertile.

majority at the 5-to 8-cell stage (36.8 percent). Blastocysts from young animals at 3½ days of pregnancy had already lost their zonae pellucidae and many could not be flushed intact from the uterus; therefore, this time was disregarded. All but four of the blastocysts flushed from senescent hamsters at 3½ days still retained zonae pellucidae and many did not exhibit the prominent blastocyst cavity which was present in all blastocysts from younger animals (3 days pregnant) flushed just before implantation. In three senescent females at 3 days and one senescent female at 3½ days, two to six blastocysts remained within the oviducts. No visible anatomical obstruction was detectable



All ova were stained with lacmoid and photographed with phase optics.

FIG. 7. An abnormal ovum from a senescent hamster 6 h after ovulation. A spermatozoon which is not visible in the photograph had penetrated the zona pellucida. The pronucleus, containing numerous nucleoli, is considerably larger than a normal pronucleus. Three polar bodies are present, but not visible in the photograph. ×500.

FIG. 8. A dispermic ovum from a senescent hamster 8 h postovulation. Both spermatozoan tails (S) are visible on the vitellus of the egg, while their heads have already formed pronuclei (Pn). ×460.

FIG. 9. A normal zygote from a senescent hamster 12 h after ovulation. Both male and female pronuclei are readily visible within the vitellus of the egg. ×500.

FIG. 10. An abnormal ovum recovered from a senescent hamster 12 h postovulation. A single spermatozoon had penetrated the ovum, but one pronucleus apparently fragmented, leaving the appearance of three pronuclei (arrows). ×500.

FIG. 11. An abnormal ovum from a senescent hamster at 12 h postovulation. A single spermatozoon had fertilized the egg, but the pronuclei are fragmented with several nucleoli (arrows) scattered throughout the ovum. ×500.

FIG. 12. An ovum from a senescent hamster recovered 12 h after ovulation. A single spermatozoon with a swollen head is present in the vitellus. This ovum was considered viable, but it is doubtful it would develop normally because of its retarded condition. ×500.

preventing transport of the blastocysts and in every female other blastocysts were recovered from the uterine horns.

Delayed Mating

To investigate whether the delay in fertilization among senescent hamsters was due to loss of spermatozoan viability resulting from their retention in the female reproductive tract, females were placed with males and observed for behavioral estrus, but removed before copulation could take place. After 4 h the female was reintroduced into a cage of males and allowed to mate. In this way spermatozoa would only have been present in the female tract for 4 to 5 h before ova entered the ampulla of the oviduct. This has been shown to be sufficient time for hamster spermatozoa to undergo capacitation (Yanagimachi and Chang, 1961; Yanagimachi, 1966). The number of ova fertilized was no greater than that of ova from females mated at the onset of estrus. Ova in which the zona pellucida was penetrated by spermatozoa at 6 h (9.3 percent) and 7 h (27.3 percent) postovulation had a lower percentage than the number penetrated at 4 h (33.9 percent) and 5 h (46.8 percent) postovulation. This percentage of penetrated ova at 6 and 7 h was also considerably lower than the percentage of ova penetrated by spermatozoa in senescent females mated at the onset of estrus (Table 2). Five senescent females, whose uterine horns were flushed at 3½ days of pregnancy, yielded only 24 normal blastocysts. This represented 4.8 blastocysts per animal, an average of two ova below the mean of 6.9 blastocysts recovered earlier. Decreasing the time spermatozoa reside in the senescent female reproductive tract did not increase the number of ova fertilized at an earlier time.

Examination of Spermatozoa

Oviducts compared from young and senescent females permitted to mate during 1 h, revealed no difference in the location or the approximate number of spermatozoa. Spermatozoa were abundant in both groups of animals at the utero-tubal junction and intramural portion of the isthmus of the oviduct becoming progressively less dense in the caudal portion of the isthmus. There were no spermatozoa detectable in the cephalic isthmus or ampulla of the oviducts of any of the animals examined during this time period.

Spermatozoa examined from the dissected oviducts of young and senescent hamsters revealed no significant difference in the number losing their acrosomes from 3 through 8 h postovulation.

Artificial Insemination

To determine how long spermatozoa of golden hamsters were capable of penetrating and fertilizing ova, young hamsters were inseminated at various times prior to the expected hour of ovulation. When females were inseminated with epididymal spermatozoa at the onset of estrus, 83.3 percent of the ova were fertilized between 8 and 9 h postovulation, and all of the ova recovered between 24 to 48 h postovulation were at the proper stage of cleavage. The percentages of ova in which the zonae pellucidae were penetrated by spermatozoa from females inseminated 14, 15, and 16 h before the expected time of ovulation were 71.1 percent, 59.2 percent and 26.3 percent. The percentages of these ova which were fertilized were 44.7 percent, 34.4 percent and 23.7 percent. All the pronuclei from fertilized ova of inseminated females at 14 and 16 h before ovulation appeared normal. However, the pronuclei of three ova at 15 h were abnormal. One ovum contained one very large pronucleus, one ovum exhibited a normal pronucleus and a fragmented pronucleus, and another ovum had both pronuclei fragmented. From females between 24 to 48 h postovulation 61.9 percent (14 h), 60.0 percent (15 h), and 34.8 percent (16 h) of the ova had cleaved, however, all but 28.6 percent, 28.0 percent, and 8.7 percent respectively had started to degenerate. In three ova at 16 h cleavage had not occurred in spite of a spermatozoon having penetrated the zona pellucida. These data indicate that inseminated spermatozoa are capable of penetrating the zona pellucida and fertilizing some ova after residing 14 to 16 h within the female reproductive tract, but that the viability of ova fertilized in this manner is greatly reduced.

DISCUSSION

The majority of ova from senescent female hamsters remain unfertilized 2 to 5 h longer than those from young females. This does not result from ovulation occurring at a later time in the aged female, although a slight delay existed when statistically comparing the ovulation rate in the two groups of animals at 1 h postovulation. However, seven or more ova were recovered from each aged female during this time period, and at 2 h there was no statistical difference between the mean number acquired from senescent hamsters compared with ova from young hamsters at 1 h postovulation. This indicates no more than a 1 h difference in the completion of ovulation between young and aged animals.

The number of ova recovered from both groups of hamsters during the first 12 h was not significantly different, although slightly fewer ova were recovered from older females compared with younger females at each time interval. These data are in agreement with other studies comparing the ovulation rate in young and senescent hamsters (Thorneycroft and Soderwall, 1969; Connors *et al.*, 1972; Blaha and Leavitt, 1974).

Some of the early ova obtained from senescent females were defective at the time of ovulation, since degenerating ova were present before fertilization would normally occur. These ova may not have been naturally ovulated, but may have been similar to degenerating ova found in females exhibiting excessive numbers of ova. Other normal appearing ova from senescent females may have been non-viable at the time of ovulation because some fertilized ova were found abnormal after staining with lacmoid. It appears likely, however, that some of these abnormalities resulted from the delay in fertilization.

Transplantation studies conducted to determine the viability of fertilized ova in senescent mice and hamsters are contradictory. Blaha (1964) transferred early embryos (63–68 h postovulation) from aged hamsters to the uteri of young pseudopregnant recipients and found the percentage approaching parturition was only 4.5 percent compared to 49.2 percent in the control group (young to young transfers). Talbert and Krohn (1966), employing similar techniques on mice, achieved results in which the number of blastocysts from old animals surviving in young females (54 percent) was essentially the same as the number surviving when transferred from young to young animals (48 percent). Gosden (1974) recently repeated the study by Talbert and Krohn achieving similar results. The differences in these studies have been explained largely as species differences; however, the delay in fertilization noted in the present study could account for some of the increased embryonic loss Blaha reported in the hamster.

It is well documented that delaying fertilization for a few hours in golden hamsters adversely affects ovum viability. Yanagimachi and Chang (1961) artificially inseminated female hamsters 3 h after the expected time of ovulation and found 14 percent of the ova infertile 8½ to 10 h later. No attempt was made to determine what percentage of fertilized ova would cleave and develop normally for implantation. In a similar study in which chromosomal anomalies were examined, Yamamoto and Ingalls (1972) found that 63 percent of the ova recovered at 3 days failed to cleave when female hamsters were delayed from mating until 3 h after the expected time of ovulation. They also found that eight of 47 embryos (17 percent) which were fertilized exhibited chromosomal aberrations. In another group of females mated in the same way, at 9 days of pregnancy they found only seven of 15 females fertile and 30 percent of the implants in a state of resorption. Of the normal implants examined for chromosomal patterns, 6 percent were abnormal. Neither of the above studies considered 2 to 4 h for spermatozoan capacitation (Yanagimachi, 1966); therefore, both studies show that normal fertilization and development are influenced by aging ova 5 to 7 h after ovulation. Many of the ova from senescent hamsters in the present study were not fertilized between 5 and 8 h postovulation. This delay would have been sufficient to help cause the high mortality rate in preimplantation embryos.

The results obtained in this study cannot be attributed to the strain of animals raised in the Oregon colony, since senescent females purchased from two other laboratory farms also exhibited a similar delay in fertilization. At every time interval examined, some senescent females from outside the Oregon colony were utilized. Seasonal conditions, likewise, failed to alter the pattern of fertilization or preimplantation loss in aged hamsters.

Miyamoto and Chang (1972) found the fertilizable life of inseminated golden hamster spermatozoa within the female reproductive tract was about 13 h. Although 2-3 percent of the ova were fertilized when insemination occurred from 13 to 15½ h before the expected time of ovulation, only 0-3 percent of the unfertilized ova had the zona pellucida penetrated by spermatozoa (Miyamoto, personal communication). In the present study a greater percentage of ova was fertilizable from young inseminated females and from senescent females (mated naturally at the onset of estrus) at a time when spermatozoa should be incapable of penetrating the zona pellucida, according to Miyamoto and Chang (1972). However, most of the ova fertilized from young females inseminated 14–16 h before the expected time of ovulation in our study either failed to cleave, or if cleavage occurred, degenerated during the first 48 h. This would imply that the viability of spermatozoa in the reproductive tract of the young female is comparable in both studies.

Blaha and Leavitt (1974) recently analyzed plasma samples from senescent hamsters (17–23 months of age) for progesterone content by gas chromatography, and found elevated levels of progesterone in some females 1 day after ovulation. The higher levels of circulating progesterone these investigators detected in some females may explain the results found in this study. The delayed fertilization exhibited by senescent hamsters could be the result of slowed spermatozoan transport. Hunter (1968) injected hamsters with varying doses of progesterone on the afternoon prior to the evening of ovulation or on the evening of ovulation, and found a reduction in the number of fertilized ova. He attributed the decline in fertility to diminished transport of spermatozoa. Chang (1967, 1969) and Chang and Hunt (1970) have also shown that progesterone and progesterone compounds administered to rabbits in low oral contraceptive doses adversely affect fertilization by speeding egg transport to the uterus and by impeding spermatozoan transport. In the present study, transport of spermatozoa through the reproductive tract, 1 h after the beginning of coitus, was the same in young and senescent animals. The absence of spermatozoa in the cephalic isthmus and ampulla of oviducts at this time agrees with that of an eariler study by Yanagimachi and Chang (1963). Although spermatozoa from aged females may be impeded in their migration through the oviduct at a later time, spermatozoa may also be prolonged in capacitation. The ability of spermatozoa to capacitate in the uterus of pseudopregnant rabbits and rabbits treated with progestins was inhibited, while the capacitation of spermatozoa in the fallopian tubes of similarly treated rabbits appeared relatively unaffected (Chang, 1958; Bedford, 1970; Brown and Hamner, 1971). Perhaps capacitation is slowed in the uterus of the senescent hamster. Bedford (1967) was able to extend the fertilizable life of motile rabbit spermatozoa by placing them in the uterus of a rat before reinsemination into a female rabbit. Since capacitation did not occur in the rat uterus, this experiment suggested that spermatozoa have a finite period of viability once capacitation is completed. This could explain why spermatozoa are capable of penetrating and fertilizing ova in senescent hamsters at a time when they have been shown to be unfertilizable in young hamsters.

The uterus is generally accepted as the organ most responsible for a declining litter size among aging laboratory animals. This appears to be true in the senescent hamster since the uterus is probably responsible for the increased number of fetal resorptions noted in older females (Thorneycroft and Soderwall, 1969). Although preimplantation loss in this study is the most important factor reducing litter size in older hamsters, this is not necessarily inconsistent with other studies. For example, it has been suggested that higher levels of circulating progesterone in aged rabbits and hamsters may result from uterine failure to respond to or bind progesterone (Larson *et al.* 1972; Blaha and Leavitt, 1974). A refractory uterus could then be indirectly responsible for delaying fertilization by elevating the level of circulating progesterone, which could slow spermatozoan transport, and more probably, prolong spermatozoan capacitation. Aging of the ovum as a result of delayed fertilization would increase the possibility of developmental anomalies leading to increased embryonic loss.

Preimplantation loss in the senescent hamster is the most important cause of litter decline in these mammals. Approximately 40 percent of the total ova produced fail to implant. The 2 to 5 h delay in fertilization of the majority of ova from aged hamsters may be the primary factor contributing to non-viable ova, although some ova appear to be inherently defective at the time of ovulation. Additional studies will have to be conducted to determine the cause of delayed fertilization in senescent golden hamsters.

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